

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks																				
Q. 1	Solve Any Two of the following.		12																				
A)	Explain in brief FSM Properties and Limitations? Write Applications of FSM?	Understand / CO-1	6																				
B)	Construct Mealy Machine for the given Moore Machine?	Apply/ CO-2	6																				
	<table border="1"> <thead> <tr> <th>Present state</th><th colspan="2">Next State</th><th>Output</th></tr> <tr> <th></th><th>a=0</th><th>a=1</th><th></th></tr> </thead> <tbody> <tr> <td>→q1</td><td>q1</td><td>q2</td><td>0</td></tr> <tr> <td>q2</td><td>q1</td><td>q3</td><td>0</td></tr> <tr> <td>q3</td><td>q1</td><td>q3</td><td>1</td></tr> </tbody> </table>	Present state	Next State		Output		a=0	a=1		→q1	q1	q2	0	q2	q1	q3	0	q3	q1	q3	1		
Present state	Next State		Output																				
	a=0	a=1																					
→q1	q1	q2	0																				
q2	q1	q3	0																				
q3	q1	q3	1																				
C)	Design a FSM to check whether a given decimal number is divisible by 5?	Understand/ CO-1	6																				
Q.2	Solve Any Two of the following.		12																				
A)	Explain in brief Chomsky hierarchy with suitable examples?	Understand / CO-2	6																				
B)	Eliminate NULL production from the grammar G given below: a. $S \rightarrow a / Xb / aYa$ b. $X \rightarrow Y / \wedge$ c. $Y \rightarrow b / X$	Apply / CO-3	6																				
	Write the productions after elimination.																						
C)	Check whether the given grammar is ambiguous or not- a. $S \rightarrow a / abSb / aAb$ b. $A \rightarrow bS / aAAb$	Analyse / CO-3	6																				

Q. 3	Solve Any Two of the following.		12
A)	Consider a left linear grammar as given below – a. $S \rightarrow Sa Abc$ b. $A \rightarrow Sa Ab a$ Convert left linear grammar to right linear grammar.	Apply / CO-3	6
B)	Convert the given to context free grammar to CNF forms a. $S \rightarrow ASB$ b. $A \rightarrow aAS a \epsilon$ c. $B \rightarrow SbS A bb$	Apply / CO-3	6
C)	Derive a derivation tree for the string aabbabba for the given context free grammar (CFG) – a. $S \rightarrow aB bA$ b. $A \rightarrow a aS bAA$ c. $B \rightarrow b bS aBB$	Analyse / CO-3	6
Q.4	Solve Any Two of the following.		12
A)	Define a PDA & list three important properties of a PDA Machine?	Understand / CO-2	6
B)	Construct a PDA to check well formedness of parenthesis?	Create / CO-3	6
C)	Convert the following grammar to a PDA that accepts the same language. a. $S \rightarrow 0S1 A$ b. $A \rightarrow 1A0 S \epsilon$	Apply / CO-3	6
Q. 5	Solve Any Two of the following.		12
A)	Design a TM for an equal number of a's and b's must follow a.	Create / CO-3	6
B)	Design a TM which recognizes palindromes over $\Sigma = \{a, b\}$	Create / CO-3	6
C)	Explain 'Halting Problem of turning machine' with neat diagrams?	Understand / CO-4	6

*** End ***